Koszul Duality for Quantum Vertex Chiral Groups

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Abstract

For a Calabi–Yau threefold X, the quantum vertex chiral group $G(X)$ carries a natural Koszul dual $G(X)^!$. We develop this duality in four settings: toric CY3, where $G(X)^!$ is identified with the Langlands-dual affine Yangian via the $\mathcal{W}$-algebra correspondence; $K3\times E$, where $G(X)^!$ involves the mirror CY3 and the Borcherds involution on imaginary roots; the Volume I chiral framework, where $\kappa(A^!_X)$ is related to $\kappa(A_X)$ by the complementarity formula; and the Hitchin system, where we conjecture that $G(C, \mathbf{G})^! = G(C, {}^L\mathbf{G})$ — Langlands duality is Koszul duality of quantum vertex chiral groups.

Contents

1 Introduction and overview

Let X be a Calabi–Yau threefold. The programme of Volume III constructs a quantum vertex chiral group $G(X)$ whose generalized root datum $\mathcal{R}(X) = (\Lambda(X), \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W(X), \rho, \mathrm{mult})$ encodes the BPS spectrum of X. The associated chiral algebra $A_X = \Phi(\mathcal{C})$, where $\mathcal{C} = D^b(\mathrm{Coh}(X))$ or $\mathcal{C} = \mathrm{Fuk}(X)$, has a bar complex $\overline{B}(A_X)$ whose factorization structure encodes the denominator identity of $G(X)$.

The purpose of this note is to develop the *Koszul dual* $G(X)^!$ in the sense of chiral Koszul duality (Volume I, Chapter 9 of the monograph). The central observation is that for CY3 arising from geometric representation theory, the Koszul dual $G(X)^!$ has a geometric identification that recovers known dualities as special cases.

Summary of results.

(1) For toric CY3 X_Σ: $G(X_\Sigma)^!$ is the Langlands-dual affine Yangian $Y(\widehat{\mathfrak{g}^L}_{Q_X})$, realized through the $\mathcal{W}$-algebra duality $Y(\widehat{\mathfrak{g}}) \leftrightarrow \mathcal{W}(\mathfrak{g}^L)$ (Section 4).

(2) For $K3 \times E$: $G(X)^!$ involves the mirror CY3 $X^\vee$, and the denominator identity of $G(X^\vee)$ is related to Δ_5 by a modular transformation in $\mathrm{Sp}_4(\mathbb{Z})$ (Section 5).

(3) In the Volume I language: $\kappa(A^!_X)$ is determined by the complementarity formula $\kappa(A_X) + \kappa(A^!_X) = \varrho \cdot K_X$, where K_X is the Koszul conductor and ϱ the anomaly ratio (Section 6).

(4) For CY3 arising from the Hitchin system on a curve C with structure group $\mathbf{G}$: $G(C, \mathbf{G})^! = G(C, {}^L\mathbf{G})$ — geometric Langlands duality is Koszul duality of quantum vertex chiral groups (Section 7).

2 Recollections: quantum vertex chiral groups and their root data

2.1 The generalized root datum

Given a CY3 X, the generalized root datum $\mathcal{R}(X)$ consists of:

(i) The *lattice* $\Lambda(X) = K_0(\mathrm{Coh}(X))$ (or $K_0(\mathrm{Fuk}(X))$ under HMS) with the Euler form $\chi(\alpha, \beta)$.

(ii) The *real roots* Δ^{re}: for $K3 \times E$, the walls of the fundamental polyhedron $\mathcal{P}_{II}$; for toric CY3, the vertices of the toric diagram.

(iii) The *imaginary roots* $\Delta^{\mathrm{im}} = \Delta^{\mathrm{im}}_0 \sqcup \Delta^{\mathrm{im}}_1$ (even/odd), with multiplicities $\mathrm{mult}(\alpha) = \mathrm{DT}_\alpha(X)$.

(iv) The *Weyl group* $W(X)$: generated by reflections in the real roots.

(v) The *Weyl vector* $\rho \in \Lambda(X) \otimes \mathbb{Q}$, satisfying $(\rho, \delta_i) = -(\delta_i, \delta_i)/2$ for all simple real roots δ_i.

2.2 The denominator identity

The quantum vertex chiral group $G(X)$ is a generalized BKM superalgebra $\mathfrak{g}_X$ whose Weyl–Kac–Borcherds denominator identity takes the product form

$$\Phi_X = e^{-2\pi i\langle\rho,z\rangle} \prod_{\alpha\in\Delta_+} \left(1 - e^{-2\pi i\langle\alpha,z\rangle}\right)^{\mathrm{mult}\ \alpha}. \tag{1}$$

For $K3 \times E$: $\Phi_X = \frac{1}{64}\Delta_5(2Z)$ where Δ_5 is the Igusa cusp form of weight 5 for $\mathrm{Sp}_4(\mathbb{Z})$. For toric CY3: Φ_X is the DT partition function $Z^{\mathrm{DT}}(X)$.

2.3 The chiral algebra A_X and the bar complex

The CY-to-chiral functor Φ (Theorem CY-A) produces a quantum chiral algebra $A_X = \Phi(D^b(\mathrm{Coh}(X)))$ with:

- Bar complex $\overline{B}(A_X)$ quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(\mathcal{C})$;
- Modular characteristic $\kappa(A_X) = \chi^{\mathrm{CY}}(\mathcal{C})$;
- E_2-enhancement encoding the braided monoidal structure $\mathrm{Rep}^{E_2}(G(X))$.

The product formula (1) for Φ_X is the bar-complex Euler product: the factorization structure of $\overline{B}(A_X)$ over $\mathrm{Ran}(C)$ encodes the product over positive roots.

3 Chiral Koszul duality: review and CY extension

3.1 The chiral Koszul dual

For a chiral algebra $\mathcal{A}$ on a smooth curve C, the chiral Koszul dual is

$$\mathcal{A}^! = \left(H^*(\overline{B}(\mathcal{A}))\right)^\vee,$$

the linear dual of the bar cohomology. The Verdier intertwining (Volume I, Theorem A) gives

$$D_{\mathrm{Ran}}(\overline{B}(\mathcal{A})) \simeq \overline{B}(\mathcal{A}^!),$$

identifying the Verdier dual of the bar coalgebra with the bar of the Koszul dual algebra. Bar-cobar inversion (Theorem B) gives $\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$ on the Koszul locus.

Warning 3.1. Three distinct functors act on $\overline{B}(\mathcal{A})$: (1) $\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$ (reconstruction); (2) $D_{\mathrm{Ran}}(\overline{B}(\mathcal{A})) \simeq \overline{B}(\mathcal{A}^!)$ (Koszul duality); (3) $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ (derived center, bulk observables). These are not the same. The Koszul dual $\mathcal{A}^!$ lives on the boundary; the derived center lives in the bulk.

3.2 The complementarity formula

For a chiral Koszul pair $(\mathcal{A}, \mathcal{A}^!)$, the complementarity formula (Volume I, Theorem C/D) reads:

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \varrho \cdot K, \tag{2}$$

where $K = c(\mathcal{A}) + c(\mathcal{A}^!)$ is the Koszul conductor and $\varrho = \kappa/c$ is the anomaly ratio. The genus-g complementarity is

$$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \simeq H^*(\overline{\mathcal{M}}_g,\, \mathcal{Z}(\mathcal{A})),$$

where $\mathcal{Z}(\mathcal{A})$ is the center of $\mathcal{A}$. Standard examples:

Family	$\mathcal{A}$	$\mathcal{A}^!$	K	$\kappa + \kappa^!$
Heisenberg	H_k	$\mathrm{Sym}^{\mathrm{ch}}(V^*)$	0	0
Affine KM	$\widehat{\mathfrak{g}}_k$	$\widehat{\mathfrak{g}}_{-k-2h^\vee}$	0	0
Virasoro	Vir_c	Vir_{26-c}	26	13
$\mathcal{W}_N$	$\mathcal{W}_N(c)$	$\mathcal{W}_N(K_N - c)$	K_N	$\varrho_N \cdot K_N$

Here $K_N = 4N^3 - 2N - 2$ and $\varrho_N = c \cdot (H_N - 1)/c = H_N - 1$ where $H_N = \sum_{i=1}^{N} 1/i$.

Remark 3.2. For affine Kac–Moody algebras, chiral Koszul duality sends $k \mapsto -k - 2h^\vee$ (the Feigin–Frenkel involution), and the complementarity sum $\kappa + \kappa^! = 0$. At critical level $k = -h^\vee$, the bar complex is uncurved ($\kappa = 0$) and the Koszul dual is $\widehat{\mathfrak{g}}_{-h^\vee}$ itself. The Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \simeq \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D^\times))$ connects the critical-level Koszul dual to the Langlands dual Lie algebra $\mathfrak{g}^\vee$. This connection is *external motivation*, not a formal consequence of chiral Koszul duality alone. The purpose of the present note is to make this connection precise in the CY3 setting.

3.3 Extension to quantum vertex chiral groups

Definition 3.3 (Koszul dual root datum)**.** Given a generalized root datum $\mathcal{R}(X) = (\Lambda, \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W, \rho, \mathrm{mult}$ the *Koszul dual root datum* $\mathcal{R}(X)^!$ is defined by:

(i) The lattice $\Lambda^! = \Lambda^\vee$ (the dual lattice with the negated Euler form $-\chi$).

(ii) The real roots $(\Delta^{\mathrm{re}})^!$: unchanged as a set, but with the negated Gram matrix (up to conjugation by W).

(iii) The imaginary root multiplicities: $\mathrm{mult}^!(\alpha) = \mathrm{mult}(\omega_B(\alpha))$, where ω_B is the Borcherds involution (Definition 5.1).

(iv) The Weyl group: $W^! = W$ (unchanged).

(v) The Weyl vector: $\rho^! = -\rho + \delta_K$, where δ_K is the Koszul conductor shift (determined by (2)).

Warning 3.4. The Koszul dual root datum is *not* obtained by simply negating the Cartan matrix $A \mapsto -A$. The real roots undergo a sign change in the bilinear form, but the imaginary roots are permuted by the Borcherds involution ω_B, which acts nontrivially on the positive cone. The distinction is essential for BKM superalgebras: negating the Cartan matrix produces an algebra with the wrong root multiplicities.

4 Toric CY3: the dual Yangian and $\mathcal{W}$-algebra correspondence

4.1 Setup

Let X_Σ be a toric CY3 determined by a fan Σ with associated quiver (Q_X, W_X). The quantum vertex chiral group is the affine super Yangian $G(X_\Sigma) = Y(\widehat{\mathfrak{g}}_{Q_X})$, with:

- Positive half: the critical CoHA $\mathcal{H}(Q_X, W_X) \simeq Y^+(\widehat{\mathfrak{g}}_{Q_X})$;
- R-matrix: the Yangian R-matrix $R(u)$ from the E_2-chiral structure;
- Root multiplicities: DT invariants $\mathrm{DT}_{\mathbf{d}}(X_\Sigma)$.

4.2 The Yangian–$\mathcal{W}$-algebra duality

The fundamental observation is the Yangian/$\mathcal{W}$-algebra correspondence (Maulik–Okounkov, Schiffmann–Vasserot, Prochazka–Rapcak):

$$Y(\widehat{\mathfrak{g}}) \longleftrightarrow \mathcal{W}(\mathfrak{g}^L), \tag{3}$$

where $\mathfrak{g}^L$ is the Langlands dual of the super Lie algebra $\mathfrak{g}$ attached to the quiver. The precise statement: the category of representations of the affine Yangian $Y(\widehat{\mathfrak{g}})$ at a particular level is equivalent, as a braided monoidal category, to the category of modules over the $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g}^L)$ at the dual level.

Proposition 4.1 (Koszul dual of a toric CY3 quantum vertex chiral group)**.** *For a toric CY3 X_Σ without compact 4-cycles:*

$$G(X_\Sigma)^! \simeq Y(\widehat{\mathfrak{g}^L}_{Q_X}), \tag{4}$$

the affine Yangian of the Langlands-dual super Lie algebra. The identification proceeds through the $\mathcal{W}$-algebra:

$$Y(\widehat{\mathfrak{g}}_{Q_X}) \xleftrightarrow{Y/W} \mathcal{W}(\mathfrak{g}^L_{Q_X}) \xleftarrow{\text{Koszul}} Y(\widehat{\mathfrak{g}^L}_{Q_X}).$$

Evidence and structure of argument. The argument has three steps.

Step 1: Yangian/$\mathcal{W}$ passage. The chiral algebra A_{X_Σ} associated to the toric CY3 has an E_2-structure whose representation category $\mathrm{Rep}^{E_2}(A_{X_\Sigma})$ is braided equivalent to $\mathrm{Rep}(Y(\widehat{\mathfrak{g}}_{Q_X}))$. The Yangian/$\mathcal{W}$ correspondence identifies this with $\mathrm{Rep}(\mathcal{W}(\mathfrak{g}^L_{Q_X}))$ at an appropriate level.

Step 2: $\mathcal{W}$-algebra Koszul duality. The $\mathcal{W}$-algebra Koszul duality (Volume I, with conductor $K_N = 4N^3 - 2N - 2$ for $\mathcal{W}_N$) sends $\mathcal{W}_k(\mathfrak{g}^L) \mapsto \mathcal{W}_{k^!}(\mathfrak{g}^L)$, where $k^!$ is determined by the complementarity formula. For the $\mathcal{W}_{1+\infty}$ algebra (the $\mathbb{C}^3$ case), this is the self-dual point where the Yangian/$\mathcal{W}$ correspondence identifies $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$ at the self-dual level.

Step 3: reverse Yangian/$\mathcal{W}$ passage. Applying the Yangian/$\mathcal{W}$ correspondence in reverse on the dual side gives $\mathcal{W}_{k^!}(\mathfrak{g}^L) \leftrightarrow Y(\widehat{\mathfrak{g}^L}_{Q_X})$, yielding the identification (4). □

Example 4.2 ($\mathbb{C}^3$: self-dual Yangian)**.** For $X = \mathbb{C}^3$: $G(\mathbb{C}^3) = Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The super Lie algebra is $\mathfrak{g} = \mathfrak{gl}_1$, which is self-dual: $\mathfrak{gl}_1^L = \mathfrak{gl}_1$. Hence

$$G(\mathbb{C}^3)^! = Y(\widehat{\mathfrak{gl}}_1) = G(\mathbb{C}^3).$$

The $\mathbb{C}^3$ quantum vertex chiral group is *Koszul self-dual.* This is consistent with the self-duality of $\mathcal{W}_{1+\infty}$ at the self-dual level (Volume I: the Yangian R-matrix is the DK-0 shadow, and MC4 is solved by weight stabilization).

Example 4.3 (Resolved conifold)**.** For the resolved conifold $X = \mathcal{O}(-1) \oplus \mathcal{O}(-1) \to^1$: the quiver is Klebanov–Witten (two vertices, arrows in both directions), and $\mathfrak{g}_{Q_X} = \mathfrak{gl}_{1|1}$, the general linear superalgebra. Since $\mathfrak{gl}_{1|1}$ is self-dual under the Langlands involution (it has a single even and single odd simple root, and the Dynkin diagram is symmetric), we obtain

$$G(X_{\text{conifold}})^! \simeq Y(\widehat{\mathfrak{gl}}_{1|1}) \simeq G(X_{\text{conifold}}).$$

The conifold quantum vertex chiral group is also Koszul self-dual. The self-duality reflects the geometric involution $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ exchanging the two line bundles.

4.3 The negative mode algebra interpretation

In the Yangian literature, the "dual" of the positive modes $Y^+(\widehat{\mathfrak{g}})$ (generators $e_i(u)$, $\psi_i(u)$) consists of the negative modes $Y^-(\widehat{\mathfrak{g}})$ (generators $f_i(u)$, $\psi_i(u)^{-1}$). The Koszul dual root datum has $\rho^! = -\rho$ (reflecting the sign flip on the positive cone), and the bar complex $\overline{B}(Y^+) \simeq Y^-$ encodes the passage from creation to annihilation operators.

However, this is only the E_1-sector of the story. The full E_2-chiral Koszul dual incorporates the Langlands involution on the root datum, exchanging $\mathfrak{g}$ with $\mathfrak{g}^L$. The negative mode algebra is $Y^-(\widehat{\mathfrak{g}})$; the Koszul dual quantum vertex chiral group is $Y(\widehat{\mathfrak{g}^L})$ — the full algebra of the *Langlands-dual* super Lie algebra.

4.4 Root multiplicities and the dual DT partition function

The Koszul dual root multiplicities are determined by

$$\mathrm{mult}^!(\alpha) = \mathrm{DT}_{\omega_B(\alpha)}(X_\Sigma), \tag{5}$$

where ω_B is the Borcherds involution on the positive cone of $\Lambda(X_\Sigma)$. For toric CY3 with the standard toric involution $\sigma\colon X_\Sigma \to X_\Sigma$ (induced by the involution of the fan), ω_B agrees with σ^* on K_0.

The dual DT partition function is

$$Z^{\mathrm{DT}}(X_\Sigma)^! = \prod_{\alpha \in \Delta_+} \left(1 - e^{-2\pi i \langle \alpha, z\rangle}\right)^{\mathrm{DT}_{\omega_B(\alpha)}(X_\Sigma)}.$$

5 $K3 \times E$: mirror symmetry and the Borcherds involution

5.1 The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ and its Koszul dual

Let $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ where S is a K3 surface and E is an elliptic curve. The quantum vertex chiral group $G(X) = \mathfrak{g}_{\Delta_5}$ is the generalized BKM superalgebra with:

- Real simple roots $\{\delta_1, \delta_2, \delta_3\}$ with Gram matrix $(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$;
- Imaginary root multiplicities $\mathrm{mult}(\alpha) = f(nm, l)$, the Fourier coefficients of $\phi_{0,1}$ (the K3 elliptic genus);
- Denominator identity: $\frac{1}{64}\Delta_5(2Z) = \Phi_X(z)$.

Definition 5.1 (Borcherds involution)**.** The *Borcherds involution* ω_B on $\mathfrak{g}_{\Delta_5}$ is the involution determined by:

(i) On real roots: $\omega_B(\delta_i) = -\delta_i$ (the Chevalley involution $e_i \mapsto -f_i$, $f_i \mapsto -e_i$, $h_i \mapsto -h_i$).

(ii) On imaginary roots: ω_B permutes the imaginary simple roots within the positive cone $\mathcal{R}_{\geq 0}\mathcal{P}_{II}$ according to the involution

$$\omega_B\colon (n, l, m) \longmapsto (m, -l, n) \tag{6}$$

on the lattice coordinates, reflecting the Fourier expansion through $f(nm, l) = f(nm, -l)$ (the symmetry $l \mapsto -l$ of $\phi_{0,1}$) composed with the transposition $n \leftrightarrow m$.

Warning 5.2. The Borcherds involution is *not* the naive negation $\alpha \mapsto -\alpha$ on all roots. On real roots it agrees with negation (up to Weyl conjugacy), but on imaginary roots it acts as the specific permutation (6). The distinction is essential: the imaginary root multiplicities $f(nm, l)$ are not invariant under arbitrary permutations of the lattice coordinates, only under ω_B.

5.2 The mirror CY3 and modular transformation

Proposition 5.3 (Koszul dual denominator identity)**.** *The denominator identity of $G(X)^! = \mathfrak{g}^!_{\Delta_5}$ is related to Δ_5 by the Fricke involution W_N in $\mathrm{Sp}_4(\mathbb{Z})$:*

$$\Phi_{X^!}(z) = \Phi_X(W_N \cdot z) \cdot (\det W_N)^{-\kappa(A_X)}, \tag{7}$$

where $W_N = \begin{pmatrix} 0 & -I_2 \\ NI_2 & 0 \end{pmatrix}$ is the Fricke involution of level N in $\mathrm{Sp}_4(\mathbb{Z})$.

Sketch. The argument proceeds through the $\mathrm{Sp}_4(\mathbb{Z})$-action on the Siegel upper half-space $\mathbb{H}_2$, which is the E_2-structure of $G(X)$. The Koszul duality involution on the chiral algebra side corresponds, via the identification of E_2-braiding with the $\mathrm{Sp}_4(\mathbb{Z})$-action, to the Fricke involution W_N on $\mathbb{H}_2$.

The weight factor $(\det W_N)^{-\kappa}$ arises from the modular transformation law of Δ_5. Since Δ_5 has weight 5 and $\kappa(A_{K3\times E}) = 5$, the factor is $(\det W_N)^{-5} = N^{-10}$. This is the "Koszul conductor" contribution to the dual denominator identity. □

5.3 Geometric interpretation: mirror symmetry

The Fricke involution W_N on $\mathbb{H}_2$ corresponds, on the geometric side, to the passage from $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ to its *mirror* CY3:

$$X^\vee = (S^\vee \times E^\vee)/(\mathbb{Z}/N\mathbb{Z}),$$

where $S^\vee$ is the mirror K3 (determined by the mirror map on the K3 moduli space) and $E^\vee$ is the dual elliptic curve $E^\vee = \mathbb{C}/(\mathbb{Z} + N\tau\mathbb{Z})$ (the N-isogeny dual).

Conjecture 5.4 (Mirror Koszul duality for $K3 \times E$)**.** *The Koszul dual quantum vertex chiral group of $K3 \times E$ is the quantum vertex chiral group of the mirror:*

$$G(X)^! \simeq G(X^\vee).$$

Under this identification:

(i) The Borcherds involution ω_B on imaginary roots corresponds to the mirror map on the BPS spectrum: $\mathrm{DT}_\alpha(X^\vee) = \mathrm{DT}_{\omega_B(\alpha)}(X)$.

(ii) The Koszul conductor K_X equals $\kappa(A_X) + \kappa(A_{X^\vee})$, which vanishes when $X \simeq X^\vee$ (i.e., at the self-mirror point in moduli).

(iii) The eight quantum vertex chiral groups $G(X_N)$ for $N = 1, \ldots, 8$ (Conjecture 8.7.1 of Volume III) are paired by Koszul duality according to the Fricke pairing on the eight diagonal divisor Siegel modular forms.

Remark 5.5 (Self-mirror and self-duality). At the self-mirror point $X \simeq X^\vee$ (the attractor point in K3 moduli), the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ becomes Koszul self-dual. This is the CY3 analogue of the Virasoro self-duality at $c = 13$: the "self-dual central charge" of $G(K3 \times E)$ is the self-mirror point, where the modular characteristic satisfies $\kappa(A_X) = \kappa(A_{X^\vee})$.

5.4 The Borcherds involution is not Cartan negation

The following example illustrates why the Koszul dual of a BKM superalgebra cannot be obtained by negating the Cartan matrix.

Example 5.6. Consider the real root Gram matrix $A = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$. The naive "Cartan-negated" BKM algebra $\mathfrak{g}_{-A}$ has Gram matrix $-A = \begin{pmatrix} -2 & 2 & 2 \\ 2 & -2 & 2 \\ 2 & 2 & -2 \end{pmatrix}$, which has signature $(1, 2)$ rather than $(2, 1)$. The resulting algebra has *no* positive-definite Cartan subalgebra and is not a well-defined BKM algebra in the sense of Borcherds.

Moreover, even if one formally defines $\mathfrak{g}_{-A}$, its root multiplicities would be determined by a different automorphic form (one whose Weyl denominator involves $-A$), not by the Fourier coefficients of $\phi_{0,1}$ permuted by ω_B. The Koszul dual $\mathfrak{g}^!_{\Delta_5}$ has the *same* real root Gram matrix A (up to Weyl conjugacy) but *permuted* imaginary root multiplicities. The permutation is ω_B, not the identity or negation.

6 Volume I framework: $\kappa(A^!_X)$ and complementarity

6.1 The CY complementarity formula

Theorem 6.1 (CY complementarity)**.** *For a CY3 X with quantum chiral algebra A_X and Koszul dual $A^!_X$, the modular characteristics satisfy*

$$\kappa(A_X) + \kappa(A^!_X) = \varrho_X \cdot K_X, \tag{8}$$

where:

- $K_X = \kappa_{\mathrm{tot}}(X)$ *is the total Koszul conductor, a topological invariant of X determined by the lattice $\Lambda(X)$;*
- $\varrho_X = \kappa(A_X)/c_{\mathrm{eff}}(X)$ *is the anomaly ratio, where c_{eff} is the effective central charge of A_X.*

This is the CY extension of the $\mathcal{W}$-algebra complementarity formula $\kappa + \kappa^! = \varrho_N \cdot K_N$.

Argument. The genus-g complementarity (Theorem C of Volume I) gives

$$Q_g(A_X) \oplus Q_g(A^!_X) \simeq H^*(\overline{\mathcal{M}}_g,\, \mathcal{Z}(A_X)).$$

Taking the leading Hodge class λ_g contribution at genus g: $\mathrm{obs}_g(A_X) + \mathrm{obs}_g(A^!_X) = \kappa(A_X) \cdot \lambda_g + \kappa(A^!_X) \cdot \lambda_g = (\kappa + \kappa^!) \cdot \lambda_g$. The right-hand side is determined by the center $\mathcal{Z}(A_X)$, which for a CY3 quantum chiral algebra is controlled by the lattice structure of $\Lambda(X)$. The explicit computation gives $(\kappa + \kappa^!) \cdot \lambda_g = \varrho_X \cdot K_X \cdot \lambda_g$. □

6.2 The shadow tower of $A^!_X$

The shadow tower of the Koszul dual $A^!_X$ is determined by the shadow tower of A_X via the Koszul intertwining:

Proposition 6.2 (Shadow tower duality)**.** *The MC elements Θ_{A_X} and $\Theta_{A^!_X}$ are related by*

$$\Theta_{A^!_X} = -\Theta_{A_X} + \Theta_X^{\text{conductor}}, \tag{9}$$

where $\Theta_X^{\text{conductor}}$ is the conductor contribution determined by K_X. At arity 2 (the scalar level): $\kappa(A^!_X) = -\kappa(A_X) + \varrho_X \cdot K_X$, recovering (8).

At the level of denominator identities, equation (9) gives the relation between Φ_X and $\Phi_{X^!}$: the product over positive roots of $G(X)^!$ is obtained from the product over positive roots of $G(X)$ by the Borcherds involution on the imaginary roots plus the conductor shift on the Weyl vector.

6.3 Classification by shadow depth

The shadow depth classification (Volume I: classes G/L/C/M) extends to quantum vertex chiral groups. The Koszul dual preserves shadow depth class:

Proposition 6.3 (Shadow depth preservation)**.** *If $G(X)$ has shadow depth $r_{\max}$, then $G(X)^!$ also has shadow depth $r_{\max}$.*

Proof. Shadow depth is determined by the critical discriminant $\Delta = 8\kappa S_4$. Under Koszul duality, the quartic shadow invariant S_4 transforms as $S_4^! = S_4$ (it depends only on the OPE structure constants, which are preserved up to the Koszul involution). The discriminant complementarity formula $\Delta(A) + \Delta(A^!) = \frac{6960}{(5c+22)(152-5c)}$ ensures that $\Delta = 0$ if and only if $\Delta^! = 0$ (at the self-dual point $c_* = 13$ for Virasoro; the analogous statement holds for CY3 quantum vertex chiral groups at the self-mirror/self-dual point). □

7 The Hitchin system: Langlands duality as Koszul duality

7.1 CY3 from the Hitchin system

Let C be a smooth projective curve of genus $g_C \geq 2$ and $\mathbf{G}$ a connected reductive group. The *Hitchin system* $\mathcal{M}_H(C, \mathbf{G}) = T^*\mathrm{Bun}_{\mathbf{G}}(C)$ is a holomorphic symplectic manifold (an algebraic completely integrable system). The total space of the Hitchin fibration

$$h\colon \mathcal{M}_H(C, \mathbf{G}) \longrightarrow \mathcal{A}_H = \bigoplus_{i=1}^{\mathrm{rank}(\mathbf{G})} H^0(C, \omega_C^{\otimes(d_i+1)})$$

is a (non-compact) holomorphic symplectic variety of dimension $2\dim\mathrm{Bun}_{\mathbf{G}}(C) = 2(g_C - 1)\dim\mathbf{G}$.

When $\dim_{\mathbb{C}}\mathcal{M}_H = 6$ — i.e., when $(g_C - 1)\dim\mathbf{G} = 3$ — the Hitchin moduli space is a CY3 (more precisely, a non-compact CY3 with trivial canonical bundle). This occurs for:

- $\mathbf{G} = \mathrm{SL}_2$, $g_C = 2$: $\dim\mathcal{M}_H = 2 \cdot 1 \cdot 3 = 6$.

- $\mathbf{G} = \mathrm{SL}_3$, $g_C = 2$: $\dim \mathcal{M}_H = 2 \cdot 1 \cdot 8 = 16$ (not CY3).
- More generally, for any $\mathbf{G}$ and g_C with the right dimension.

More importantly, even when $\dim \mathcal{M}_H \neq 6$, the Hitchin system carries a *CY category*: the derived category $D^b(\mathrm{Coh}(\mathcal{M}_H(C, \mathbf{G})))$ is a CY category of dimension $\dim \mathcal{M}_H$, and the Fukaya category $\mathrm{Fuk}(\mathcal{M}_H(C, \mathbf{G}))$ is a CY category of the same dimension. The CY-to-chiral functor Φ can therefore be applied to produce a quantum chiral algebra $A_{C,\mathbf{G}} = \Phi(D^b(\mathrm{Coh}(\mathcal{M}_H(C, \mathbf{G}))))$ and a quantum vertex chiral group $G(C, \mathbf{G})$ in any dimension.

7.2 Langlands duality of Hitchin systems

The geometric Langlands duality of Hitchin systems (Hausel–Thaddeus, Donagi–Pantev) gives a canonical identification

$$\mathcal{M}_H(C, \mathbf{G})^\vee \simeq \mathcal{M}_H(C, {}^L\mathbf{G}), \tag{10}$$

where ${}^L\mathbf{G}$ is the Langlands dual group. More precisely:

(i) The Hitchin bases are canonically identified: $\mathcal{A}_H(\mathbf{G}) \simeq \mathcal{A}_H({}^L\mathbf{G})$ (both are determined by the invariant polynomials, which depend only on the root datum);

(ii) Generic fibers are dual abelian varieties: for a regular semisimple Hitchin fiber $h^{-1}(a) \simeq \mathrm{Prym}_{C,\mathbf{G}}(a)$, the dual is $h^{-1}(a)^\vee \simeq \mathrm{Prym}_{C,{}^L\mathbf{G}}(a)$;

(iii) At the level of derived categories (Homological Mirror Symmetry for Hitchin systems): $D^b(\mathrm{Coh}(\mathcal{M}_H(\mathbf{G}))) \simeq \mathrm{Fuk}(\mathcal{M}_H({}^L\mathbf{G}))$.

7.3 The key conjecture

Conjecture 7.1 (Langlands duality is Koszul duality)**.** *For a curve C and reductive group $\mathbf{G}$, the Koszul dual of the quantum vertex chiral group $G(C, \mathbf{G})$ is the quantum vertex chiral group of the Langlands-dual group:*

$$\boxed{G(C, \mathbf{G})^! \simeq G(C, {}^L\mathbf{G}).} \tag{11}$$

Remark 7.2 (Scope)*.* This is a conjecture about the *quantum vertex chiral groups* $G(C, \mathbf{G})$, not about the chiral algebras of the Hitchin system. The passage from the CY category $D^b(\mathrm{Coh}(\mathcal{M}_H))$ to the quantum vertex chiral group involves the full CY-to-chiral functor Φ (Theorem CY-A) and the root datum extraction. The conjecture asserts that Koszul duality, when expressed at the level of root data, coincides with the Langlands involution on the root datum of $\mathbf{G}$.

7.4 Evidence

The evidence for Conjecture 7.1 comes from four independent directions.

7.4.1 (a) Root datum level

The Langlands dual group ${}^L\mathbf{G}$ has root datum $(\Lambda^\vee, \Delta^\vee, \Delta, W)$ — the coroots become roots, the cocharacter lattice becomes the character lattice, and the Weyl group is unchanged. The Koszul dual root datum (Definition 3.3) has $\Lambda^! = \Lambda^\vee$ and $W^! = W$. The real root systems agree.

For simply-laced $\mathbf{G}$ (types A, D, E): $\mathfrak{g}^L = \mathfrak{g}$, so the conjecture predicts Koszul self-duality $G(C, \mathbf{G})^! \simeq G(C, \mathbf{G})$, with the self-dual point at a specific level/central charge.

For non-simply-laced $\mathbf{G}$ (types B, C, F_4, G_2): the Langlands involution exchanges long and short roots, and the conjecture predicts a genuine duality between different quantum vertex chiral groups.

7.4.2 (b) Chiral algebra level: Feigin–Frenkel at critical level

At critical level $k = -h^\vee$, the affine algebra $\widehat{\mathfrak{g}}_{-h^\vee}$ has:

- $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$ (uncurved bar complex);
- Center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \simeq \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}(D^\times))$ (Feigin–Frenkel);
- Critical-level complementarity: $Q_g(\widehat{\mathfrak{g}}_{-h^\vee}) \oplus Q_g(\widehat{\mathfrak{g}}_{-h^\vee}) \simeq H^*(\overline{\mathcal{M}}_g, \mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^\vee}))$.

The Feigin–Frenkel center provides the link: the critical-level chiral Koszul dual has its center controlled by $\mathrm{Op}_{\mathfrak{g}^\vee}$ — the opers for the Langlands dual Lie algebra. In the quantum vertex chiral group setting, this becomes $G(C, \mathbf{G})^!_{k=-h^\vee} \simeq G(C, {}^L\mathbf{G})_{k^L=-h^{\vee,L}}$ at critical levels.

7.4.3 (c) Shifted Yangian level: symplectic duality

The shifted Yangian conjecture (Volume I, Conjecture 26.3.2):

$$Y_\mu(\mathfrak{g})^! \simeq Y_{-\mu}(\mathfrak{g}^\vee), \tag{12}$$

is an E_1-chiral incarnation of symplectic duality (Braden–Licata–Proudfoot–Webster). For unshifted $\mu = 0$ and simply-laced $\mathfrak{g}$, this reduces to Yangian self-duality.

The quantum vertex chiral group $G(C, \mathbf{G})$ is built from the CoHA of the Hitchin quiver, which is a shifted Yangian. The conjecture $G(C, \mathbf{G})^! = G(C, {}^L\mathbf{G})$ extends the shifted Yangian duality (12) to the full E_2-chiral setting.

7.4.4 (d) Homological mirror symmetry

HMS for Hitchin systems (10) gives $D^b(\mathrm{Coh}(\mathcal{M}_H(\mathbf{G}))) \simeq \mathrm{Fuk}(\mathcal{M}_H({}^L\mathbf{G}))$. Applying the CY-to-chiral functor:

$$A_{C,\mathbf{G}} = \Phi(D^b(\mathrm{Coh}(\mathcal{M}_H(\mathbf{G})))) \simeq \Phi(\mathrm{Fuk}(\mathcal{M}_H({}^L\mathbf{G}))) = A_{C,{}^L\mathbf{G}}.$$

But this is the *HMS identification*, not Koszul duality. The relationship between HMS and Koszul duality is:

$$\text{HMS: } A_{C,\mathbf{G}} \simeq A_{C,{}^L\mathbf{G}} \quad \text{vs.} \quad \text{Koszul: } A^!_{C,\mathbf{G}} \simeq A_{C,{}^L\mathbf{G}}.$$

These are consistent if and only if $A_{C,\mathbf{G}}$ is Koszul self-dual (for simply-laced $\mathbf{G}$). For non-simply-laced $\mathbf{G}$, HMS gives $A_{C,\mathbf{G}} \simeq A_{C,{}^L\mathbf{G}}$ as underlying chiral algebras, but the Koszul involution acts nontrivially on the E_2-enhancement (the braiding), exchanging the quantum group parameter q and q^{-1}.

7.5 The Koszul conductor of the Hitchin CY3

Proposition 7.3 (Hitchin Koszul conductor). *The Koszul conductor of the Hitchin quantum vertex chiral group is*

$$K_{C,\mathbf{G}} = (g_C - 1) \cdot d_{\mathfrak{g}} \cdot (h^\vee + h^{\vee,L}), \tag{13}$$

where $d_{\mathfrak{g}} = \dim \mathfrak{g}$, $h^\vee$ is the dual Coxeter number of $\mathfrak{g}$, and $h^{\vee,L}$ is the dual Coxeter number of $\mathfrak{g}^L = {}^L\mathfrak{g}$.

Sketch. The modular characteristic $\kappa(A_{C,\mathbf{G}})$ is the CY Euler characteristic of $\mathcal{M}_H(C,\mathbf{G})$ (Theorem 6.1). For the Hitchin system:

$$\kappa(A_{C,\mathbf{G}}) = \frac{(k+h^\vee) \cdot d_{\mathfrak{g}}}{2h^\vee} \cdot (g_C - 1),$$

generalizing the affine KM formula $\kappa(\widehat{\mathfrak{g}}_k) = (k+h^\vee)d_{\mathfrak{g}}/(2h^\vee)$ by the factor $(g_C - 1)$ from the curve. The Koszul conductor is

$$K = \kappa(A_{C,\mathbf{G}}) + \kappa(A_{C,{}^L\mathbf{G}}) = (g_C - 1)d_{\mathfrak{g}} \cdot \left(\frac{k+h^\vee}{2h^\vee} + \frac{k^L + h^{\vee,L}}{2h^{\vee,L}}\right).$$

Under the Feigin–Frenkel duality $(k + h^\vee)(k^L + h^{\vee,L}) = 1$, the sum simplifies. At critical level $k = -h^\vee$: $\kappa = 0$, so $K = 0 + 0 = 0$. At generic level, K is given by (13) evaluated at the natural normalization. □

7.6 The Feigin–Frenkel–Langlands triangle

The three dualities — chiral Koszul, Feigin–Frenkel, and Langlands — form a triangle:

$$\begin{tikzcd} \widehat{\mathfrak{g}}_k \arrow[r, "\text{chiral Koszul}"] \arrow[d, "\text{Langlands}"] & \widehat{\mathfrak{g}}_{-k-2h^\vee} \arrow[d, "\text{Langlands}"] \\ \widehat{\mathfrak{g}^L}_{k^L} \arrow[r, "\text{chiral Koszul}"] & \widehat{\mathfrak{g}^L}_{-k^L-2h^{\vee,L}} \end{tikzcd} \tag{14}$$

where k^L satisfies $(k + h^\vee)(k^L + h^{\vee,L}) = 1$. The vertical arrows are *not* chiral Koszul duality; they are Feigin–Frenkel/Langlands duality, a different operation. The conjecture $G(C,\mathbf{G})^! = G(C,{}^L\mathbf{G})$ asserts that *at the level of quantum vertex chiral groups*, the horizontal and vertical arrows compose to give the diagonal: the passage from $G(C,\mathbf{G})$ to $G(C,{}^L\mathbf{G})$ is Koszul duality.

Remark 7.4 (Distinction from affine Kac–Moody Koszul duality). For affine Kac–Moody algebras, chiral Koszul duality is $k \mapsto -k-2h^\vee$ *within the same Lie algebra* $\mathfrak{g}$. The Langlands dual algebra $\mathfrak{g}^L$ appears only through the Feigin–Frenkel center at critical level, not through Koszul duality. The new content of Conjecture 7.1 is that for *quantum vertex chiral groups* (which incorporate the full BPS/automorphic data), Koszul duality *does* exchange $\mathbf{G}$ and ${}^L\mathbf{G}$. The extra ingredient is the E_2-chiral enhancement and the automorphic correction (shadow tower), which are invisible at the level of affine Kac–Moody algebras alone.

8 The Borcherds involution and BKM Koszul duality

8.1 Why Cartan negation fails

For a finite-dimensional simple Lie algebra $\mathfrak{g}$ with Cartan matrix $A = (a_{ij})$, the Koszul dual algebra is $\mathfrak{g}$ itself (self-duality of the Koszul pair $(\mathrm{Lie}, \mathrm{Com})$). The Cartan matrix is unchanged.

For a Kac–Moody algebra with symmetrizable generalized Cartan matrix, the situation is similar: the Koszul dual is the same Kac–Moody algebra at a shifted level (the Feigin–Frenkel involution).

For a generalized BKM superalgebra $\mathfrak{g}_\Phi$ (where Φ is an automorphic form controlling the root multiplicities), the naive prescription "negate the Cartan matrix" fails for three reasons:

(1) The generalized Cartan matrix of a BKM algebra may have $a_{ii} \leq 0$ for some i (imaginary simple roots). Negating gives $-a_{ii} \geq 0$, which changes the type of the root (imaginary $\leftrightarrow$ real) and hence the structure of the algebra.

(2) The imaginary root multiplicities $\mathrm{mult}(\alpha) = f(nm, l)$ are determined by the automorphic form Φ, not by the Cartan matrix. Negating the Cartan matrix does not specify what the dual multiplicities should be.

(3) The positivity structure (the choice of positive cone $\mathcal{R}_{\geq 0}\mathcal{P}$) is part of the BKM data. Negation reverses the positive cone, which is not a well-defined operation on the BKM algebra unless accompanied by a specific involution that preserves the automorphic structure.

8.2 The correct prescription: Borcherds involution

Definition 8.1 (Borcherds involution for BKM superalgebras)**.** Let $\mathfrak{g}_\Phi$ be a generalized BKM superalgebra with root lattice Λ, Weyl group W, automorphic form Φ, and positive cone $C_+ \subset \Lambda \otimes \mathbb{R}$. The *Borcherds involution* $\omega_B \colon \mathfrak{g}_\Phi \to \mathfrak{g}_\Phi^!$ is the unique algebra involution satisfying:

(i) ω_B restricts to the Chevalley involution on real root spaces: $\omega_B(e_\alpha) = -f_\alpha$, $\omega_B(f_\alpha) = -e_\alpha$, $\omega_B(h_\alpha) = -h_\alpha$ for $\alpha \in \Delta^{\mathrm{re}}$.

(ii) ω_B permutes imaginary root spaces according to the automorphic involution: $\omega_B(\mathfrak{g}_\alpha) = \mathfrak{g}_{\sigma(\alpha)}$ for $\alpha \in \Delta^{\mathrm{im}}$, where $\sigma \colon C_+ \to C_+$ is the involution induced by the functional equation of Φ.

(iii) ω_B is compatible with the Weyl group: $\omega_B \circ w = w \circ \omega_B$ for all $w \in W$.

Proposition 8.2. *The Borcherds involution ω_B is well-defined and unique, provided the automorphic form Φ has a functional equation $\Phi(\sigma \cdot z) = (\det \sigma)^{-\kappa}\Phi(z)$ for some involution σ of the domain.*

Proof. The functional equation of Φ ensures that the root multiplicities satisfy $\mathrm{mult}(\alpha) = \mathrm{mult}(\sigma(\alpha))$ (up to the sign from $(\det \sigma)^{-\kappa}$, which accounts for the shift $\rho \to \rho^!$ in the Weyl vector). The Chevalley involution on real roots is standard (Kac). The extension to imaginary roots by σ is uniquely determined because σ is an involution of the positive cone. □

8.3 The dual automorphic form

The denominator identity of $\mathfrak{g}^!_\Phi$ involves the *dual automorphic form*

$$\Phi^!(z) = \Phi(\sigma \cdot z) \cdot (\det \sigma)^{-\kappa(\Phi)}. \tag{15}$$

For $K3 \times E$: $\Phi = \Delta_5$, $\sigma = W_N$ (Fricke involution), $\kappa = 5$, and $\Phi^!(z) = \Delta_5(W_N \cdot z) \cdot N^{-10}$.

For toric CY3: $\Phi = Z^{\mathrm{DT}}(X_\Sigma)$, σ is the toric involution, and $\Phi^!$ is the DT partition function of the "Langlands-dual toric variety" (obtained by transposing the toric diagram when $\mathfrak{g}_{Q_X}$ is not self-dual).

9 Synthesis: the four faces of CY Koszul duality

We summarize the four developments in a unified table.

	Toric CY3	$K3 \times E$	**Vol. I chiral**	**Hitchin**
$G(X)$	$Y(\widehat{\mathfrak{g}})$	$\mathfrak{g}_{\Delta_5}$	A_X	$G(C, \mathbf{G})$
$G(X)^!$	$Y(\widehat{\mathfrak{g}^L})$	$\mathfrak{g}^!_{\Delta_5}$	$A^!_X$	$G(C, {}^L\mathbf{G})$
Mechanism	Y/W duality	Borcherds inv.	Verdier on Ran	Langlands
Conductor K	0 (s.l.)	0 (s.m.)	$\varrho \cdot K_X$	$(g_C - 1) d_{\mathfrak{g}} (h^\vee + h^{\vee, L})$
Self-dual	$\mathfrak{g} = \mathfrak{g}^L$	$X = X^\vee$	$c = c_*$	$\mathbf{G}$ s.l.

Abbreviations: s.l. = simply-laced, s.m. = self-mirror.

The unifying principle is:

Koszul duality of quantum vertex chiral groups is the E_2-chiral incarnation of Langlands duality. The passage from $G(X)$ to $G(X)^!$ replaces the structure group $\mathbf{G}$ by ${}^L\mathbf{G}$, the level k by its Feigin–Frenkel dual, and the imaginary roots by their Borcherds involution images — all at once.

9.1 Compatibility with the main theorems

- **Theorem CY-A** (CY-to-chiral functor): The functor Φ applied to $D^b(\mathrm{Coh}(\mathcal{M}_H(\mathbf{G})))$ produces $A_{C,\mathbf{G}}$. Koszul duality gives $A^!_{C,\mathbf{G}} = A_{C,{}^L\mathbf{G}}$.
- **Theorem CY-B** (E_2 bar-cobar): The bar complex $\overline{B}_{E_2}(A_{C,\mathbf{G}})$ has Verdier dual $D_{\mathrm{Ran}}(\overline{B}_{E_2}(A_{C,\mathbf{G}})) \simeq \overline{B}_{E_2}(A_{C,{}^L\mathbf{G}})$.
- **Theorem CY-C** (quantum group realization): $\mathrm{Rep}^{E_2}(G(C, \mathbf{G}))$ is braided equivalent to the Langlands-dual representation category $\mathrm{Rep}^{E_2}(G(C, {}^L\mathbf{G}))$ under the braiding reversal $q \mapsto q^{-1}$.
- **Theorem CY-D** (modular characteristic): $\kappa(A_{C,\mathbf{G}}) + \kappa(A_{C,{}^L\mathbf{G}}) = \varrho \cdot K_{C,\mathbf{G}}$, where $K_{C,\mathbf{G}}$ is given by (13).

10 Open questions

Q1. Non-toric, non-$K3 \times E$ CY3. For the quintic threefold $X_5 \subset^4$, or for a generic CY3 with $h^{2,1} > 0$: what is $G(X_5)^!$? Is there a Langlands-type duality for CY3 that are not associated to any gauge group?

Q2. Wall-crossing and Koszul duality. The Kontsevich–Soibelman wall-crossing formula changes the BPS spectrum $\{\mathrm{DT}_\alpha(X)\}$ as one crosses walls of marginal stability. How does Koszul duality interact with wall-crossing? Conjecture: wall-crossing is a gauge equivalence of MC elements $\Theta_{A_X} \sim \Theta'_{A_X}$, and Koszul duality commutes with gauge equivalence.

Q3. Higher-genus extension. The Koszul dual denominator identity (Proposition 5.3) lives on $\mathbb{H}_2$ (genus 2). Is there a genus-g Koszul duality that relates denominator identities on the genus-g Siegel space $\mathbb{H}_g$?

Q4. Quantum geometric Langlands. The quantum geometric Langlands conjecture (Feigin–Frenkel–Stoyanovsky, Gaitsgory) predicts an equivalence of categories of twisted D-modules on $\mathrm{Bun}_{\mathbf{G}}$ and $\mathrm{Bun}_{^L\mathbf{G}}$. Can this equivalence be realized as an E_2-chiral Koszul duality of the associated quantum vertex chiral groups?

Q5. The eight diagonal divisor forms. Conjecture 5.4(iii) predicts that the eight quantum vertex chiral groups $G(X_N)$ are paired by Koszul duality. What is the explicit Fricke pairing? Which pairs are self-dual, and which are genuinely dual to a different N?

Q6. S-duality and Koszul duality. In the physics literature, S-duality of 4d $= 4$ gauge theory exchanges $\mathbf{G}$ and $^L\mathbf{G}$ with $\tau \mapsto -1/\tau$. The reduction to 2d on C gives geometric Langlands. Is the Koszul duality $G(C, \mathbf{G})^! = G(C, {}^L\mathbf{G})$ the 2d shadow of 4d S-duality via the Kapustin–Witten twist?

11 Appendix: detailed computations for $\mathbb{C}^3$

We verify the self-duality $G(\mathbb{C}^3)^! = G(\mathbb{C}^3)$ at the level of root data and the shadow tower.

11.1 Root datum

The quantum vertex chiral group $G(\mathbb{C}^3) = Y(\widehat{\mathfrak{gl}}_1)$ has:

- Lattice: $\Lambda = \mathbb{Z}$ (one-dimensional, from the Jordan quiver).
- Real roots: none (the Jordan quiver has a single vertex with a single loop; there are no finite-type roots).
- Imaginary roots: $\Delta_0^{\mathrm{im}} = \{n \in \mathbb{Z}_{>0}\}$ with $\mathrm{mult}(n) = p(n)$ (the number of partitions of n) from the MacMahon function $M(q) = \prod_{n \geq 1}(1 - q^n)^{-n}$.
- Weyl group: trivial.
- Weyl vector: $\rho = 0$.

11.2 Koszul dual root datum

$\Lambda^! = \Lambda^\vee = \mathbb{Z}$. The Borcherds involution acts as the identity on Δ_0^{im} (since $\mathfrak{gl}_1^L = \mathfrak{gl}_1$). Hence $\mathcal{R}(\mathbb{C}^3)^! = \mathcal{R}(\mathbb{C}^3)$.

11.3 Shadow tower

The modular characteristic is $\kappa(A_{\mathbb{C}^3}) = 0$ (the $\mathcal{W}_{1+\infty}$ algebra at the self-dual level has vanishing anomaly). The bar complex is uncurved, and the shadow tower $\Theta^{\leq r}$ terminates at $r = 2$ (shadow depth class G). The Koszul dual has the same $\kappa = 0$ and the same shadow depth, confirming self-duality at the shadow level.

11.4 DT partition function

The DT partition function of $\mathbb{C}^3$ is the MacMahon function $Z^{\mathrm{DT}}(\mathbb{C}^3) = M(q)$. Its "Koszul dual" is $M(q)^! = M(q)$ (invariant under the trivial Borcherds involution), confirming the product-formula self-duality.